dEZignLink for STAAD.Pro Application Usage

Designed to help Structural Engineers work more efficiently with Bentley STAAD.Pro
Featuring a collection of carefully curated tools, this toolbar delivers exceptional time savings.



Key Features

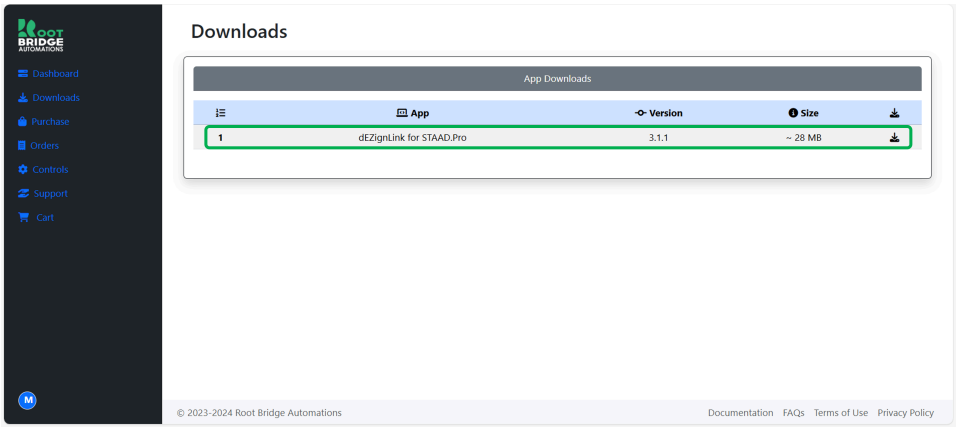
- **Select specific element types** auto-defined by dEZignLink

- **Renumber large structures** effortlessly
- **Automatically create Groups** for easy model controlling
- **Define Releases** quickly and easily
- **Assign Design Parameters** in just seconds
- **Generate essential Output Tables**, along with pivot tables
- **Generate Load Case & Combination tables** seamlessly
- **Create user-defined Plate Girders**
- **Generate Structure Geometries** in seconds

Enhance your workflow with **dEZignLink** and experience unmatched efficiency.

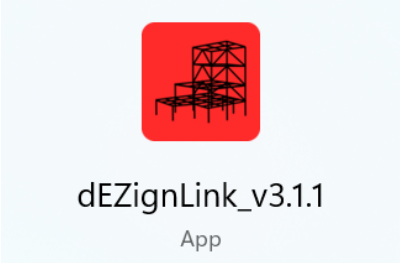
Installation

Downloaded the installer package from your account dashboard's Downloads section, as shown below:



WARNING: Installer package obtained from any other source must not be used.

Upon successful installation, shortcuts will be created in the Desktop as well as the Start Menu.



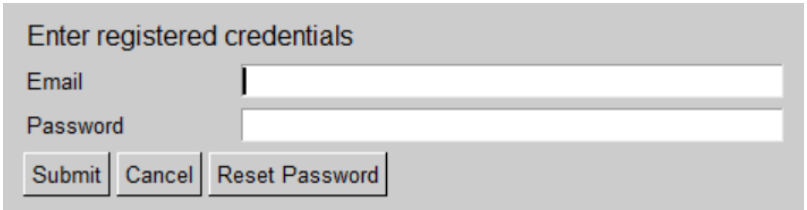
GUI Log In

After launching the tool, you need to Log In first to start using the tool.

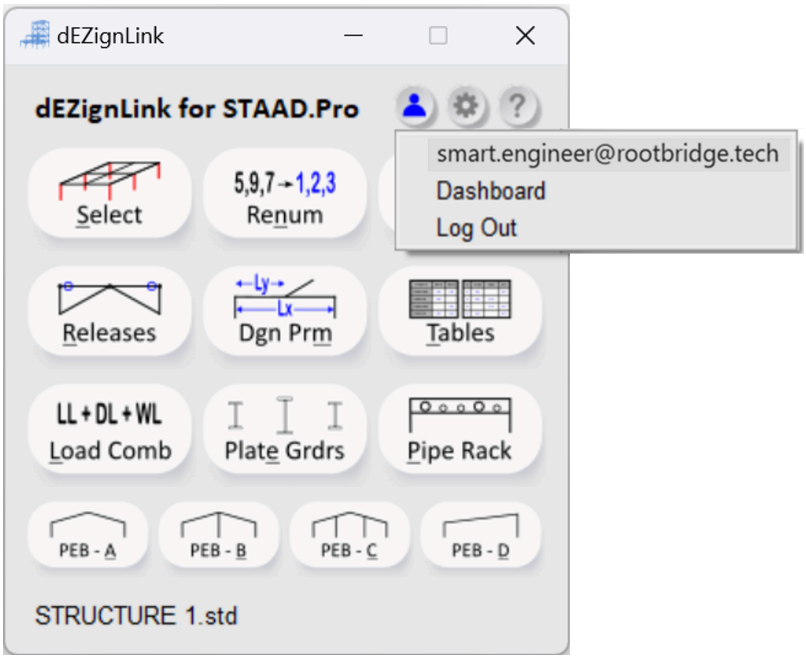
- Begin by clicking the "**Log In**" button as shown below:



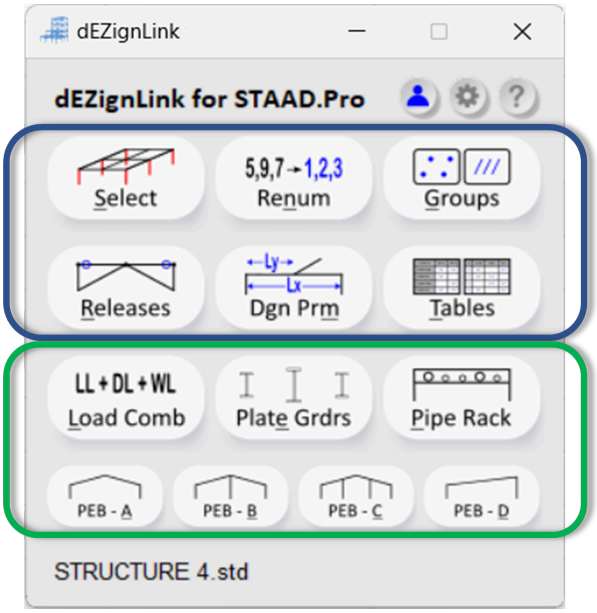
- A window will popup as shown below. Enter the same login credentials that you used to register in our website.



- Click on the "**Submit**" button. Once your login is successful, the Log In button on the tool shall change as shown below:



- Now you can use the tool. Availability of the functions shall depend upon your subscription validity. As shown below, the six tools at the top are free to use during the trial period. To continue using them after the trial, a subscription is required. All other tools remain free forever.



- The file name of the connected STAAD model is displayed at the bottom of the tool.

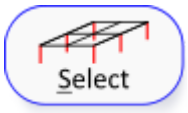
Info

OpenSTAAD can connect only to one STAAD model at a time. When dEZignLink is started, the last open STAAD model shall get connected. If you wish to connect to another model, then either of the following steps shall achieve it:
1. Close the currently connected STAAD model, before or after opening the model to which connection is required, or
2. Restart dEZignLink after opening the model to which connection is required.

Info

The session lifetime is of 30 days, after which you will have to login again.

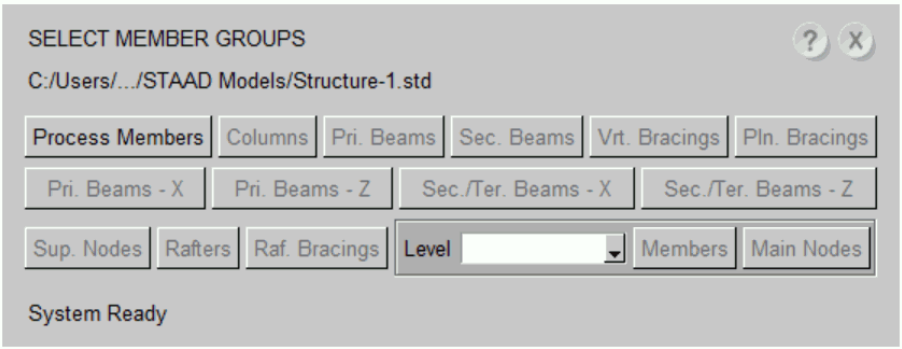
Select Members



Keyboard shortcut: Alt+S

This tool allows you to select various structural elements with a single click.

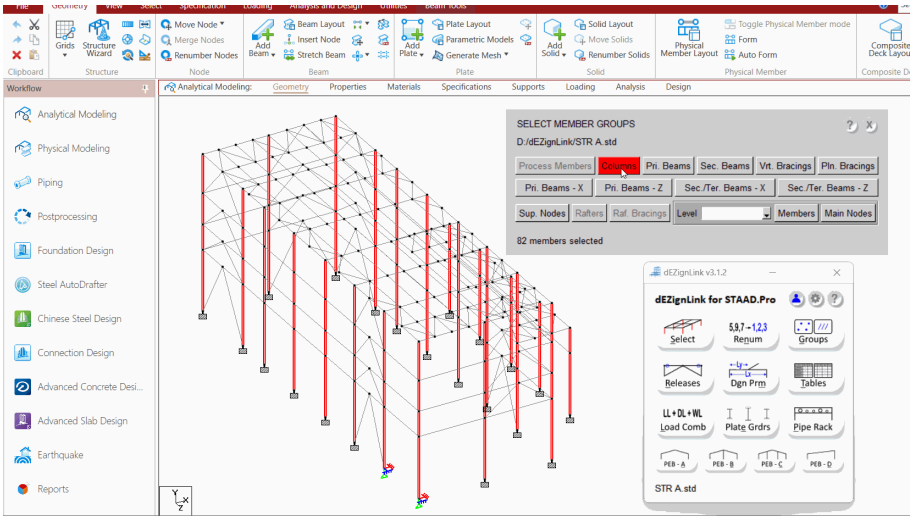
- Begin by clicking the "**Process Members**" button. This process may take a few seconds. Once completed, all other selection buttons will be activated.



- Some buttons in the SELECT tool shall remain inactive if the corresponding elements are not present in the current STAAD model. For example, if

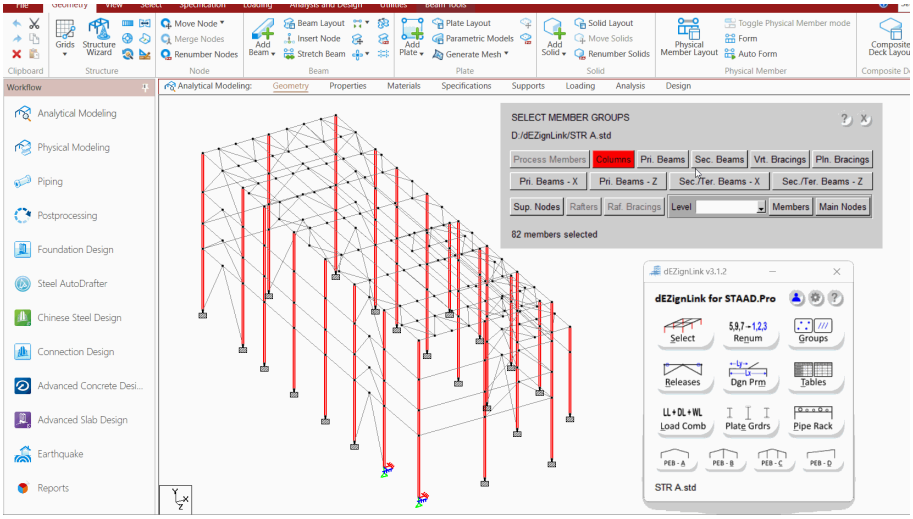
Vertical Bracings are absent in the model, the "**Vrt. Bracings**" button will not be activated.

- You can now select the required elements from your model by clicking the corresponding buttons. For example, clicking "**Pri. Beams - Z**" will select all primary beams in the current model that are parallel to the Global Z direction. (Here, primary beams refer to beams connected from column to column.)



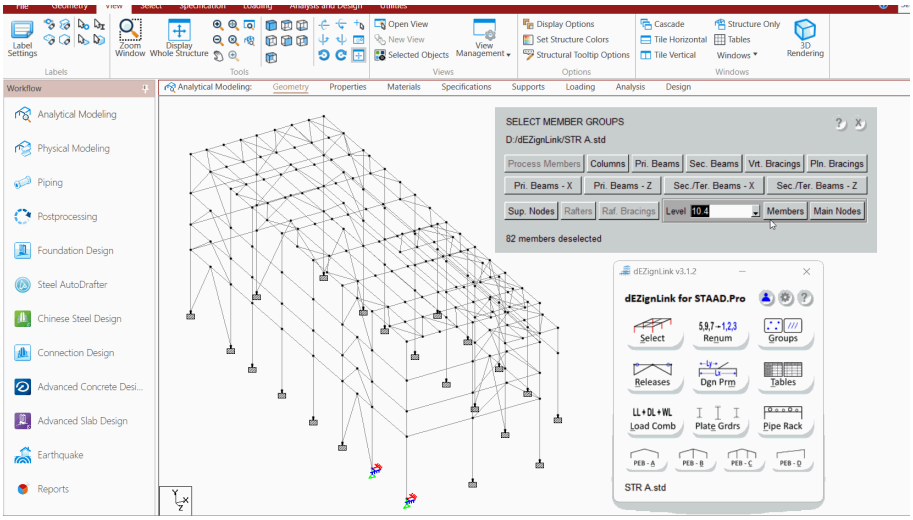
- Similarly, you can select Vertical or Plan bracings by clicking their respective buttons.

- Multiple selection buttons can be activated at the same time. When a button is selected, it will turn red. To deselect an item, click the button again; it will revert to its original colour.



- Note: If you click in the graphics window at any point, your entire selection will be cleared, although the buttons will remain red.

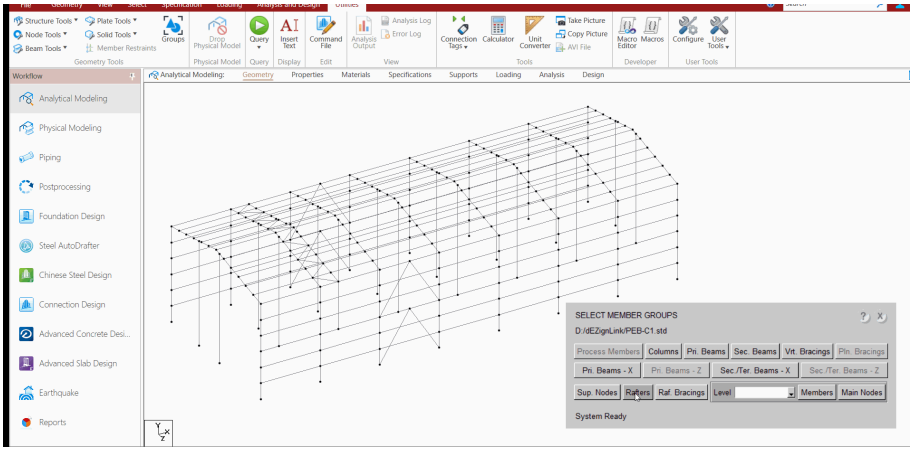
- For the "**Level**" drop down menu, choose the desired level (automatically extracted from the model) and press the "**Members**" or "**Main Nodes**" button as needed.



- Note: There are many nodes at particular level. We are generally interested in selecting Column Nodes only at each level for deflection checks and other such important functions. Here, Main Nodes refers to nodes lying on the columns.

- The buttons for "**Rafters**" or "**Rafter Bracings**" will be active only if they are present in the current model.

- In this context, Rafters refer to the inclined roof beams (primary beams) that connect the columns. Rafter bracings are the Plan bracings located within the inclined roof plane.



- If you make any geometrical changes in the model, you must press PROCESS MEMBERS again to use the SELECT Tool.

Check our video guide on Select Members

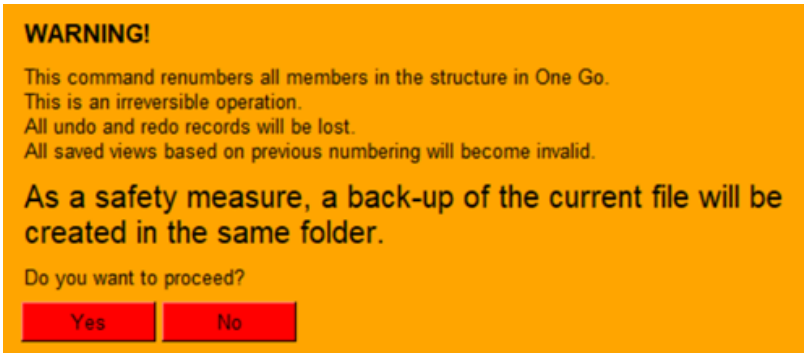
Renumbering



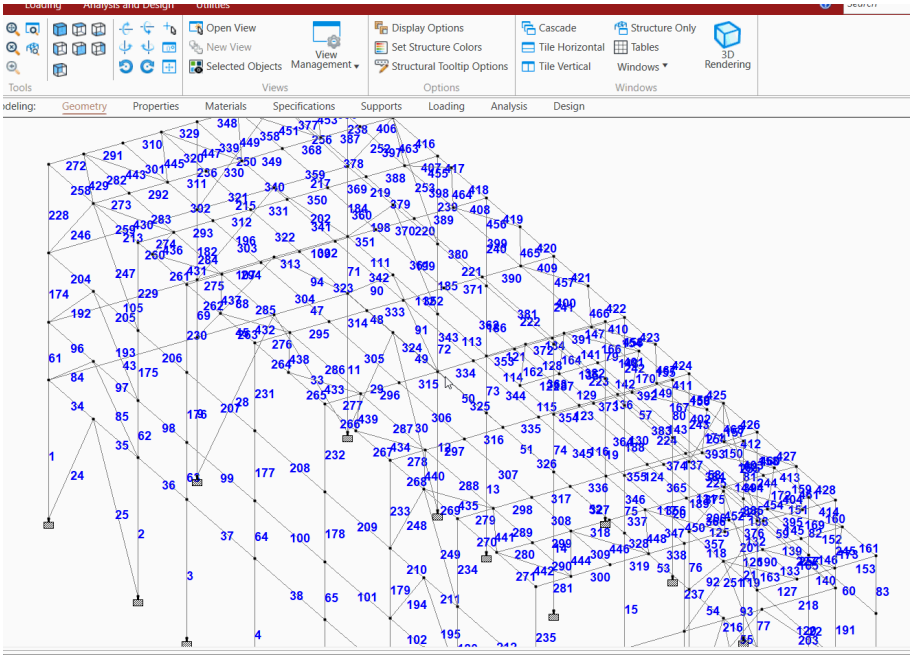
Keyboard shortcut: Alt+N

This tool effortlessly renumbers large structures with a single click.

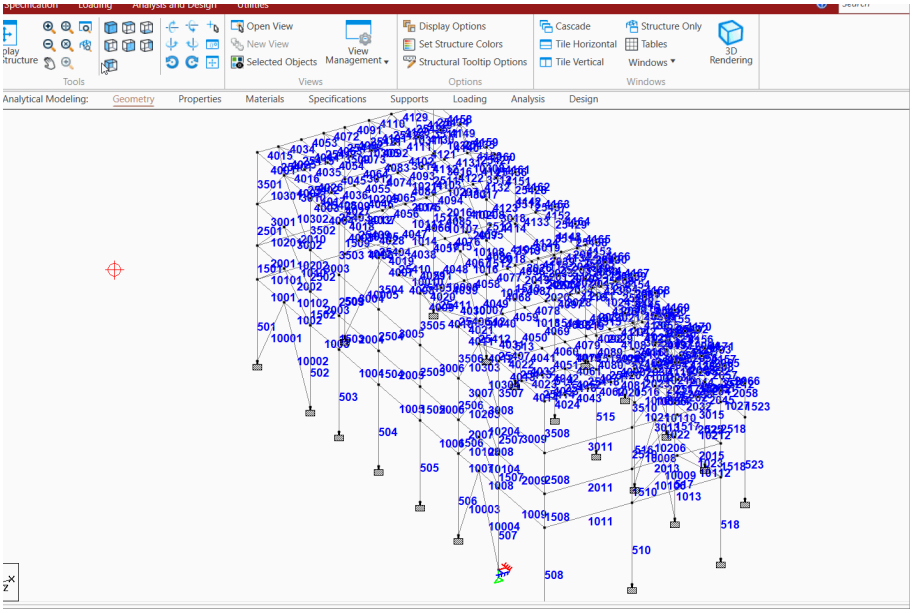
- Begin by clicking the "**Renum**" button. It will pop-up the following dialogue.



- Click "**Yes**" to proceed.
- The tool will automatically categorize elements such as Floor beams, Columns, and Bracings, and assign sequential numbering.
- For example,
    - The first floor beams will be numbered starting from 101 or 1001, while the second floor will begin from 201 or 2001, and so on.
    - Columns will be numbered starting from 1501, with subsequent floors following at 2501, etc.
    - Bracings will also be renumbered in a similar sequential pattern.
- **Before Renumbering:**



- **After Renumbering:**



- As a safety measure, a backup of the current file before renumbering is created in the same folder as the original.

Check our video guide on Renumbering

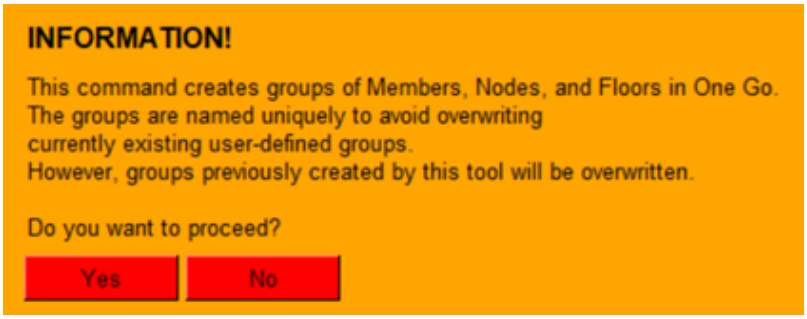
Define Groups



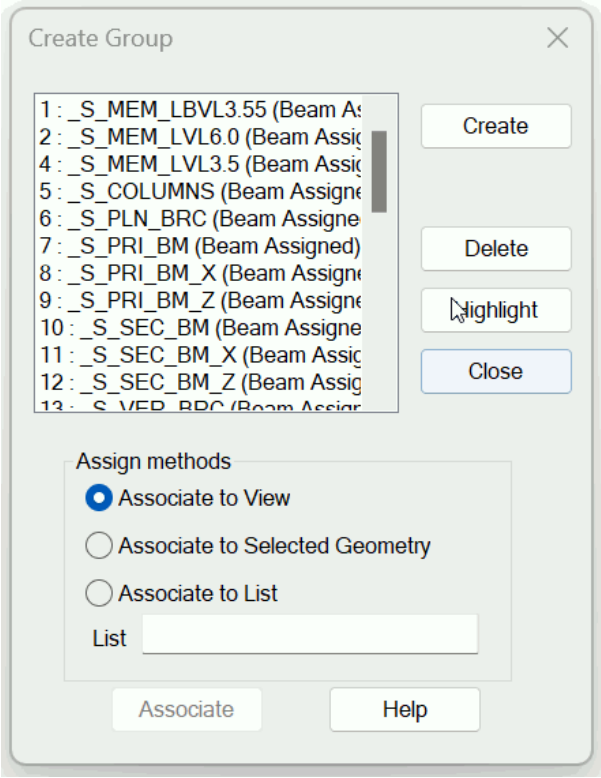
Keyboard shortcut: Alt+G

You can create multiple predefined groups for the entire structure by simply pressing this button.

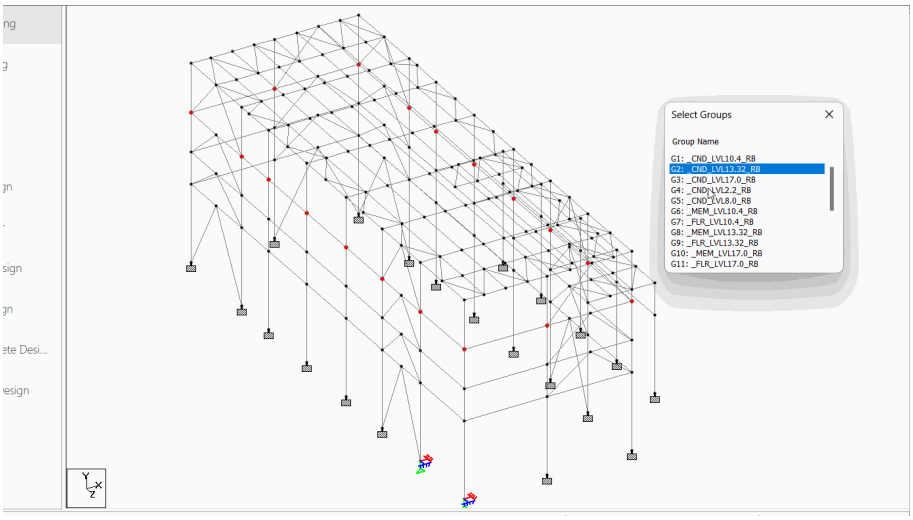
- Begin by clicking the "**Groups**" button. It will pop-up the following dialogue.



- Click "**Yes**" to proceed.
- The software will automatically generate many useful groups such as Main Beams, Secondary Beams, Floor wise Beams, Columns, Bracings, and more.
- The groups are named uniquely to avoid overwriting currently existing user-defined groups. The same can be accessed from STAAD.Pro > Utilities > Groups as shown below.



- The created groups can be selected as shown below:



- However, the groups previously created by this tool, if any, shall be overwritten.

Check our video guide on Define Groups

`The Legend used by the tool for creating GROUPS is as follows:`

Group Nomenclature	Member Types in Group
`CND`	Column Nodes
`MEM`	Members
`FLR`	Floor groups for One way - Two way loading
`PLN_BRC`	All Plan Bracings
`PRI_BM`	All Primary Beams
`SEC_TER_BM`	All Secondary & Tertiary Beams
`VER_BRC`	All Vertical Bracings

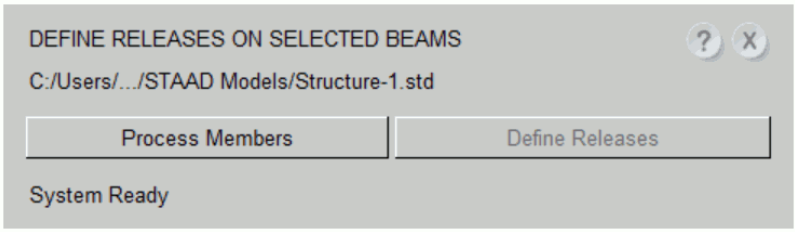
Define Releases



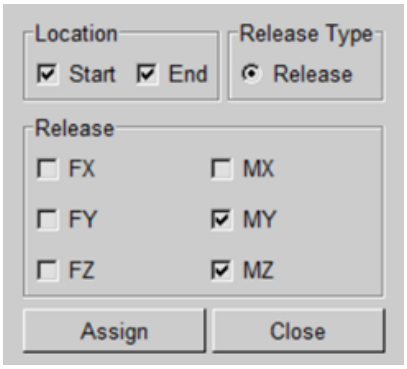
Keyboard shortcut: Alt+R

You can assign Releases (both START and END) simultaneously with this tool.

- Select all beams for which you want to define the same type of release. (Here you can use our SELECT toolbar to accelerate your work).
- Click the "**Releases**" button from the Main Toolbar. This will open a window as shown below.
- Click "**Process Members**" once to activate the "Define Releases" button.



- Next, click the "**Define Releases**" button, which will open the following window.



- In this window, both START and END releases are selected by default. Choose the type of release you wish to assign (For e.g. MY & MZ).
- Thereafter, click "**Assign**" and that's it!. The tool is designed to intelligently detect the connection location requirements for most cases. It will automatically assign releases to the:

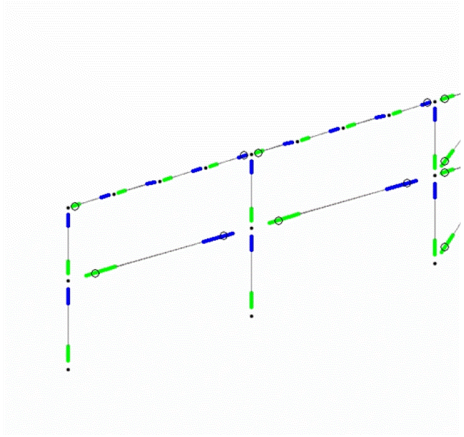
  a. start only,

  b. end only,

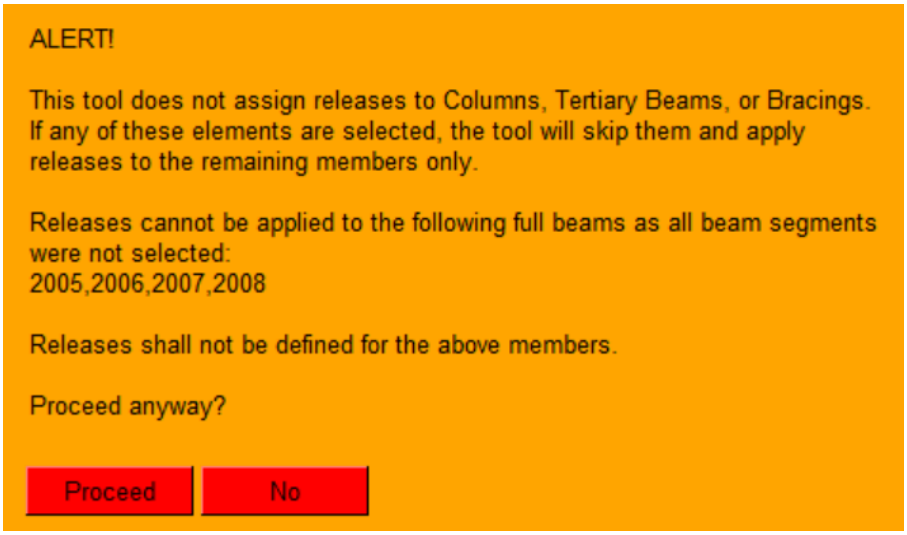
  c. both, or

  d. none,

  as applicable. It will also skip intermediate segments if necessary.

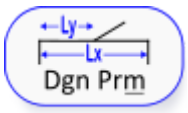


- This tool works for Primary and Secondary Beams only. If your selection contains any Columns, Bracings or, Tertiary Beams, it will show pop-up the ALERT as shown below. Also, Primary / Secondary beams whose all beam segments were not selected shall be ignored as indicated in the ALERT. In such case, you can click "**Proceed**" which will ignore the invalid type of members or incomplete selections of Primary/Secondary beams and complete the task.



Check our video guide on Releases

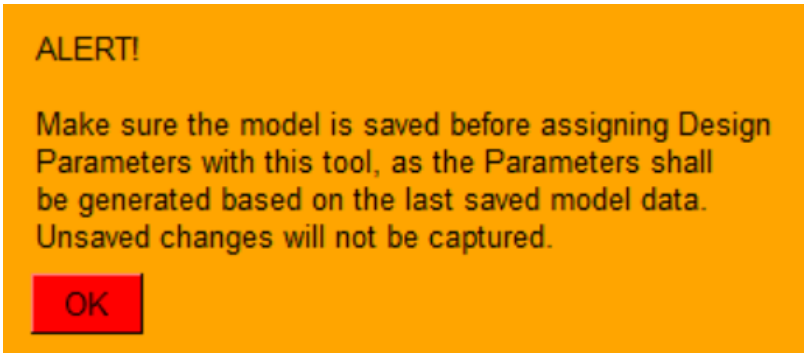
Design Parameters Generator



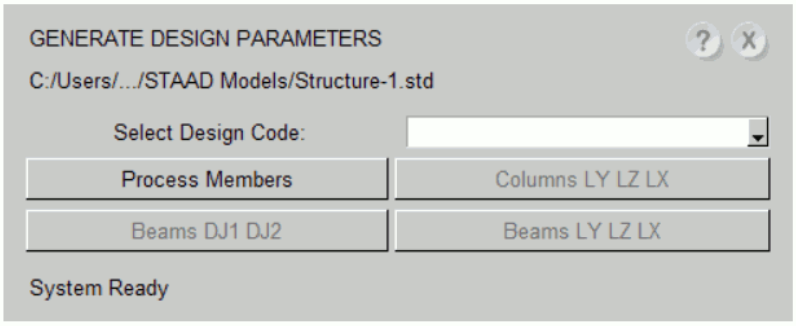
Keyboard shortcut: Alt+M

You can assign Design Parameters in compliance to various codes, with just a few clicks through this tool.

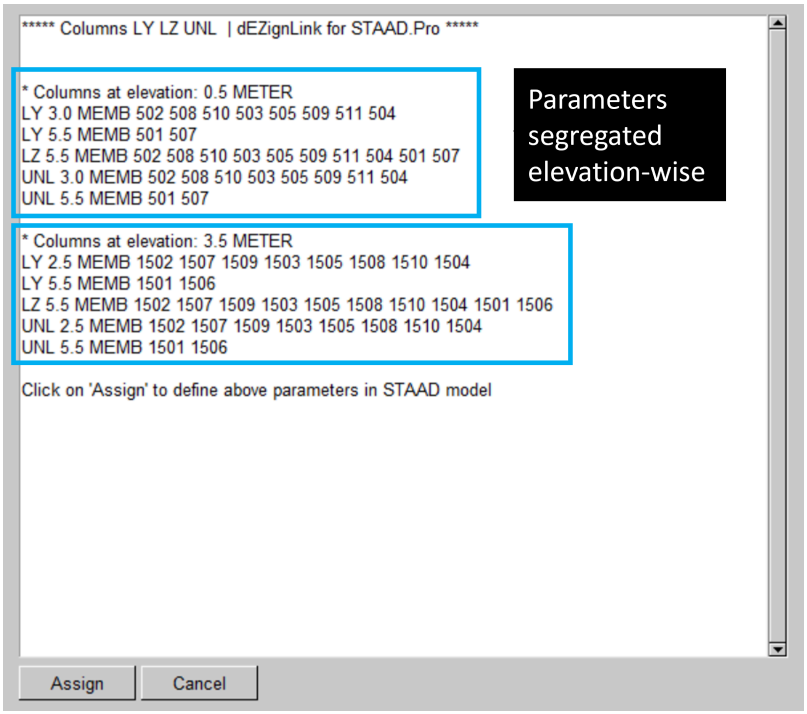
- To start with, click the "**Dgn Prm**" button. It will pop-up an alert as shown below.



- Click "**OK**" to proceed. It will open a window as shown below.
- There is no need to select individual members, in fact, no need to select anything.
- Select the desired code from the drop-down list of available codes, and click "**Process Members**".



- This action will activate related buttons for design parameters - Effective Lengths for Columns, Effective Lengths for Beams and Deflection Joints.
- Click these three buttons one by one to get the required Design Parameters.
- This process will not only define each parameter but also organize them with appropriate titles for each section.



- In order to directly apply the generated parameters to the current STAAD model, click "**Assign**", or copy the generated parameter and use it as needed.

Check our video guide on Design Parameters

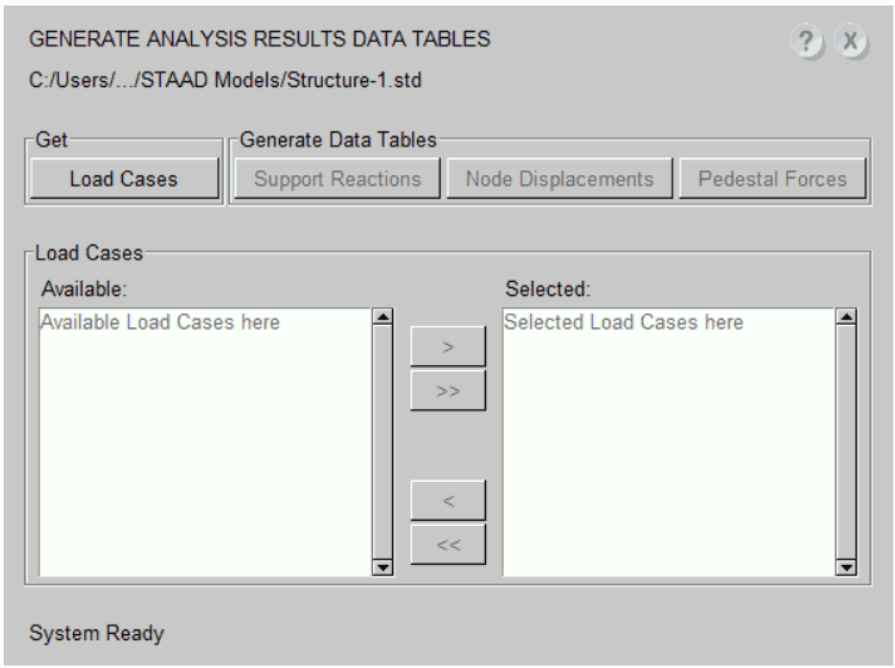
Analysis Results Tables



Keyboard shortcut: Alt+T

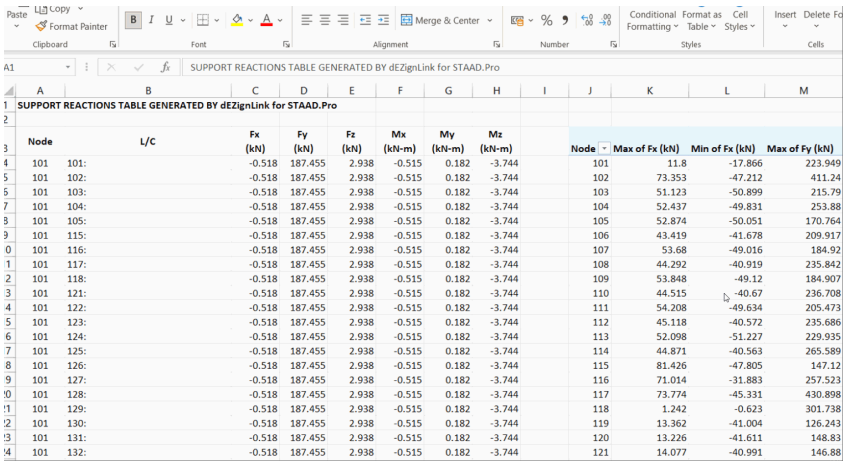
This tool provides you the Result outputs in an Excel table format, along with pivot table functionality, with just a few clicks.

- NOTE: This tool can be used only after completing the model analysis.
- Click the "**Tables**" button to open the window as shown below.
- Click "**Load Cases**" to display a list of Primary Load Cases and Load Combinations defined in your model.
- Select the required Load Cases or Load Combinations.



- Press one of three output buttons for the specific Output Table you need.

  - **Support Reactions**: This option provides an Excel file showcasing Support Reactions, with an accompanying pivot table for a clearer overview, depicting Maximum and Minimum values, as shown in the example below.

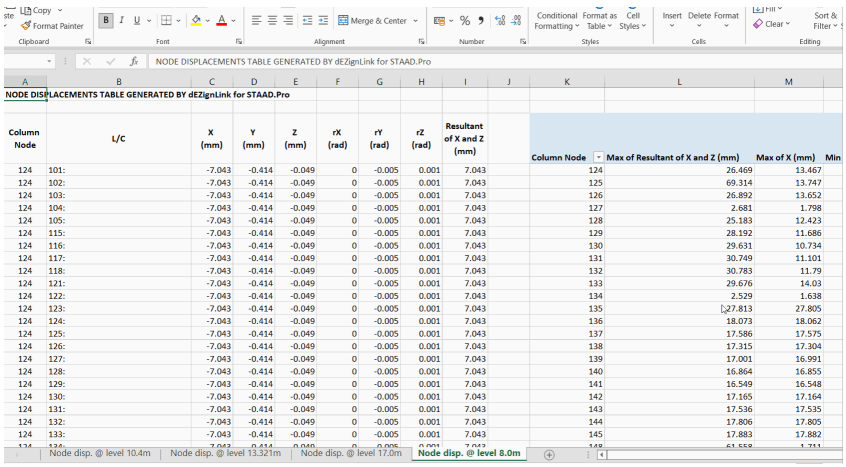


SUPPORT REACTIONS TABLE GENERATED BY dEZignLink for STAAD.Pro

Node	L/C	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
101	101:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	102:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	103:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	104:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	105:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	115:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	116:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	117:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	118:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	121:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	122:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	123:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	124:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	125:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	126:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	127:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	128:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	129:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	130:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	131:	-0.518	187.455	2.938	-0.515	0.182	-3.744
101	132:	-0.518	187.455	2.938	-0.515	0.182	-3.744

Node	Max of Fx (kN)	Min of Fx (kN)	Max of Fy (kN)
101	11.8	-17.866	223.949
102	73.353	-47.212	411.24
103	51.123	-50.899	215.79
104	52.437	-49.831	253.88
105	52.874	-50.051	170.764
106	43.419	-41.678	209.917
107	53.68	-49.016	184.92
108	44.292	-40.919	235.842
109	53.848	-49.12	184.907
110	44.515	-40.67	236.708
111	54.208	-49.634	205.473
112	45.118	-40.572	235.686
113	52.098	-51.227	229.935
114	44.871	-40.563	265.589
115	81.426	-47.805	147.12
116	71.014	-31.883	257.523
117	73.774	-45.331	430.898
118	1.242	-0.623	301.738
119	13.362	-41.004	126.243
120	13.226	-41.611	148.83
121	14.077	-40.991	146.88

  - **Node Displacements**: This feature creates an Excel file that organizes different floor levels into separate worksheets, generates Pivot tables for each level, and displays the maximum resultant displacement—all in one step. A sample screenshot is shown below.



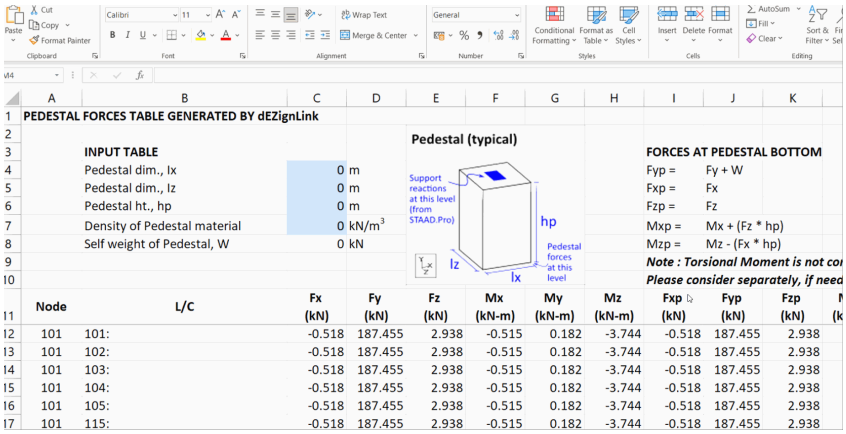
NODE DISPLACEMENTS TABLE GENERATED BY dEZignLink for STAAD.Pro

Column Node	L/C	X (mm)	Y (mm)	Z (mm)	rX (rad)	rY (rad)	rZ (rad)	Resultant of X and Z (mm)
124	101:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	102:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	103:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	104:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	105:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	115:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	116:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	117:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	118:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	121:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	122:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	123:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	124:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	125:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	126:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	127:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	128:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	129:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	130:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	131:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	132:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043
124	133:	-7.043	-0.414	-0.049	0	-0.005	0.001	7.043

Column Node	Max of Resultant of X and Z (mm)	Max of X (mm)	Min
124	26.469	13.467	
125	69.314	13.747	
126	26.892	13.652	
127	2.681	1.798	
128	25.183	12.423	
129	28.192	11.686	
130	29.631	10.734	
131	30.749	11.101	
132	30.783	11.79	
133	29.676	14.03	
134	2.529	1.638	
135	27.813	27.805	
136	18.073	18.062	
137	17.586	17.575	
138	17.315	17.304	
139	17.001	16.991	
140	16.864	16.855	
141	16.549	16.548	
142	17.165	17.164	
143	17.536	17.535	
144	17.806	17.805	
145	17.883	17.882	

Node disp. @ level 10.4m | Node disp. @ level 13.321m | Node disp. @ level 17.0m | Node disp. @ level 8.0m

- **Pedestal Forces**:
  - This tool auto-calculates the forces required for Pedestal design. The generated Excel file allows you to input specific Pedestal parameters, such as dimensions, height and density, and outputs the necessary Design forces.
  - It calculates moments by multiplying Horizontal Reactions (Fx, Fz) with height and calculates vertical forces by adding the self-weight to the Vertical reactions, providing Final Design Forces and Moments at the critical bottom section of the pedestal.
  - This is particularly useful for designing RCC Pedestals that support Steel structures where you have not modelled Pedestals.



PEDESTAL FORCES TABLE GENERATED BY dEZignLink

INPUT TABLE

Pedestal dim., lx	0	m
Pedestal dim., lz	0	m
Pedestal ht., hp	0	m
Density of Pedestal material	0	kN/m³
Self weight of Pedestal, W	0	kN

Pedestal (typical)

FORCES AT PEDESTAL BOTTOM

Fyp =	Fy + W
Fxp =	Fx
Fzp =	Fz
Mxp =	Mx + (Fz * hp)
Mzp =	Mz - (Fx * hp)

Note : Torsional Moment is not co
Please consider separately, if need

Node	L/C	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)	Fxp (kN)	Fyp (kN)	Fzp (kN)
101	101:	-0.518	187.455	2.938	-0.515	0.182	-3.744	-0.518	187.455	2.938
101	102:	-0.518	187.455	2.938	-0.515	0.182	-3.744	-0.518	187.455	2.938
101	103:	-0.518	187.455	2.938	-0.515	0.182	-3.744	-0.518	187.455	2.938
101	104:	-0.518	187.455	2.938	-0.515	0.182	-3.744	-0.518	187.455	2.938
101	105:	-0.518	187.455	2.938	-0.515	0.182	-3.744	-0.518	187.455	2.938
101	115:	-0.518	187.455	2.938	-0.515	0.182	-3.744	-0.518	187.455	2.938

- All outputs will be generated in spreadsheet format compatible with MS Excel, and saved at the default location: Desktop\dEZignLink for STAAD.Pro. You can change this path by clicking the Settings icon at the top-right corner of the dEZignLink toolbar.

Check our video guide on Results Tables

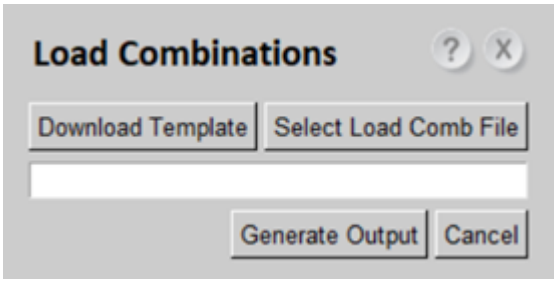
Load Combinations Generator



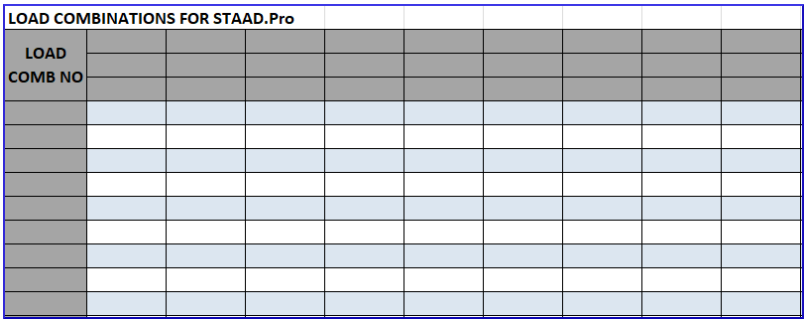
Keyboard shortcut: Alt+L

With this tool, you can generate a STAAD-compatible Load Combination file as well as a Load Case file effortlessly. Additionally, you can also get well-formatted tables for the same to use in your calculation reports.

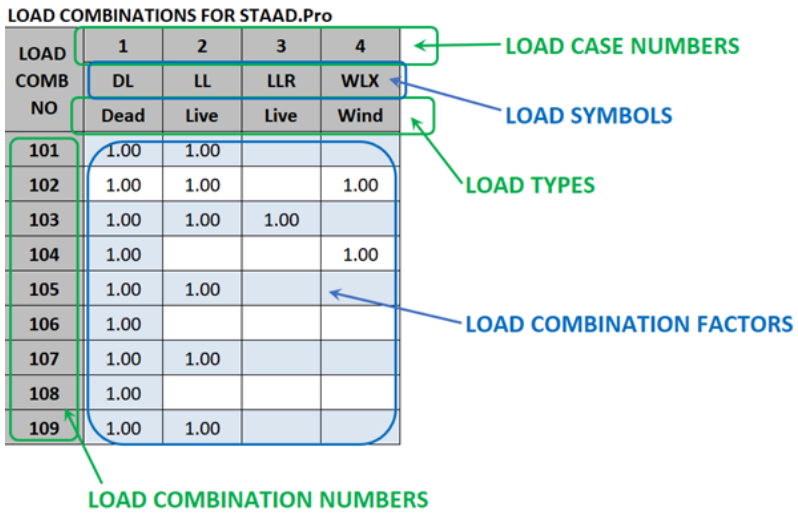
- Click the **"Load Comb"** button to open the window as shown below.
- Click the **"Download Template"** button to download the Microsift Excel compatible template for entering Load Combination data.



- Enter the Load Combination inputs in the downloaded template.



- Ensure that Load Case Numbers, Load Symbols, Load Types and Load Combination numbers are entered in the designated cells only, as shown in the screenshot below.



- Once you have completed entering all required data, save the template file at your preferred location.
- Click "**Select Load Comb File**" in the window and select the path of your filled in template file.
- Click "**Generate Output**" to create the output file.
- This tool will generate a Word file with the following content:
    - STAAD.Pro compatible Load Case script, ready for use in the STAAD Editor.

```
***** LOAD CASE DEFINITIONS | dEZignLink for STAAD.Pro *****

* STAAD Editor command lines for defining Loads.

* NOTE FOR USER:
* Copy the below command lines and paste in STAAD Editor
* below all Geometry and Specification assignment,
* specifically below "SUPPORTS" block.

LOAD 1 : [D] - Dead Load
*
LOAD 2 : [Lf] - Live Load
*
LOAD 3 : [Lr] - Roof Live Load
*
LOAD 4 : [Lp] - Roof Live Load
*
LOAD 5 : [Pt] - Snow Load
*
LOAD 6 : [Po] - Fluids Load
*
```

    - STAAD.Pro compatible Load Combination script, ready for use in the STAAD Editor.

```
***** LOAD COMBINATIONS | dEZignLink for STAAD.Pro *****

* STAAD Editor command lines for defining Load Combinations.

* NOTE FOR USER:
* Copy the below command lines and paste in STAAD Editor
* at the end of Load Case Definitions.

LOAD COMB 101  DL + LL
1 1.0 2 1.0
*
LOAD COMB 102 DL + LL+WLX+TL
1 1.0 2 1.0 4 1.0 8 1.0
*
LOAD COMB 103 DL + LL+LLR+WLZ
1 1.0 2 1.0 3 1.0 5 1.0
*
LOAD COMB 104 DL + LL+ WLZ
1 1.0 2 1.0 5 1.0
```

- Well-formatted tables for Load Cases that can be copied into your calculation reports.

LOAD NO.	LOAD SYMBOL	LOAD DESCRIPTION
1	DL	Dead Load
2	LL	Live Load
3	LLR	Live Load
4	WLX	Wind Load
5	WLZ	Wind Load
6	EQX	Seismic-H Load
7	EQZ	Seismic-H Load
8	TL	None

- Well-formatted tables for Load Combinations that can be copied into your calculation reports.

LOAD COMB NO	1	2	3	4	5	6	7	8
	DL	LL	LLR	WLX	WLZ	EQX	EQZ	TL
	Dead Load	Live Load	Live Load	Wind Load	Wind Load	Seismic-H Load	Seismic-H Load	None
101	1.0	1.0						
102	1.0	1.0		1.0				1.0
103	1.0	1.0	1.0		1.0			
104	1.0			1.0				
105	1.0	1.0			1.0			
106	1.0					1.0		
107	1.0	1.0					1.0	
108	1.0					1.0		
109	1.0	1.0					1.0	1.0

Check our video guide on Load Combinations

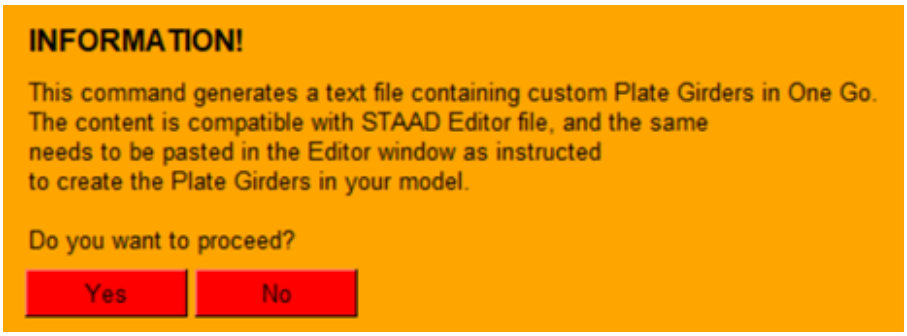
Custom Plate Girders



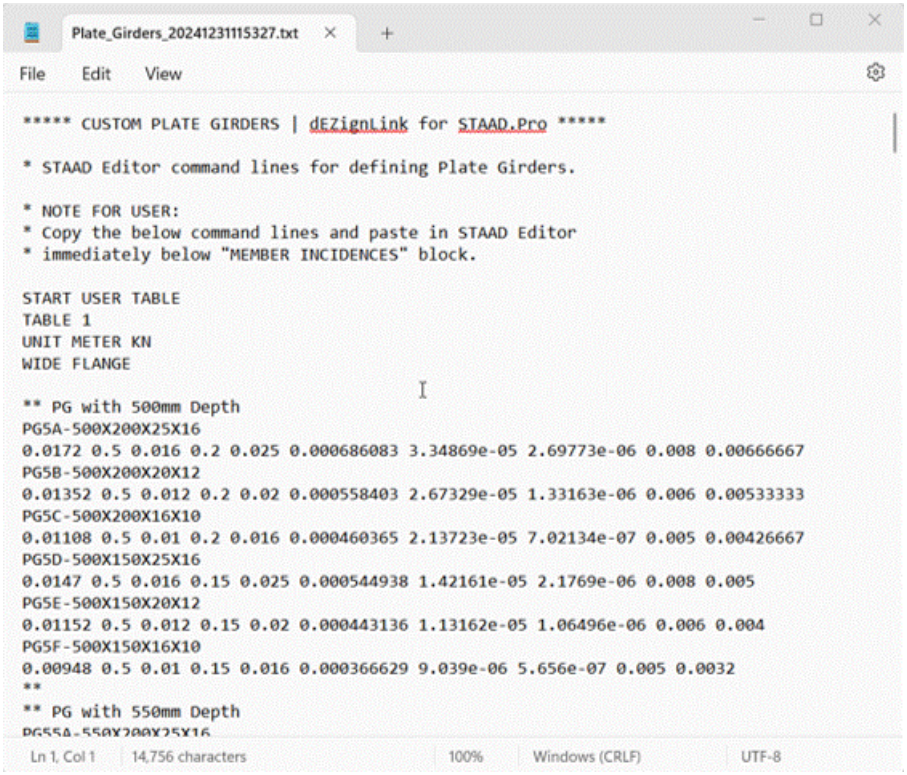
Keyboard shortcut: Alt+E

With just one click, you can access multiple pre-defined I-section plate girders, ready for use.

- Click the "**Plate Girders**" button. An information window shall appear as shown below.



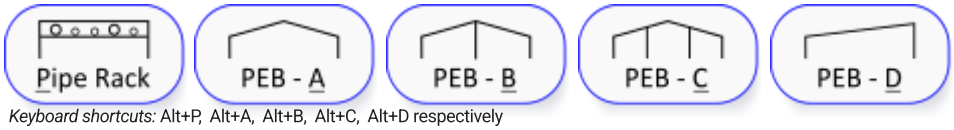
- Click "**Yes**" to proceed. This will create a text file containing a range of pre-defined plate girders.



- You can easily copy and incorporate these girders into any STAAD file as needed.

Check our video guide on Plate Girders

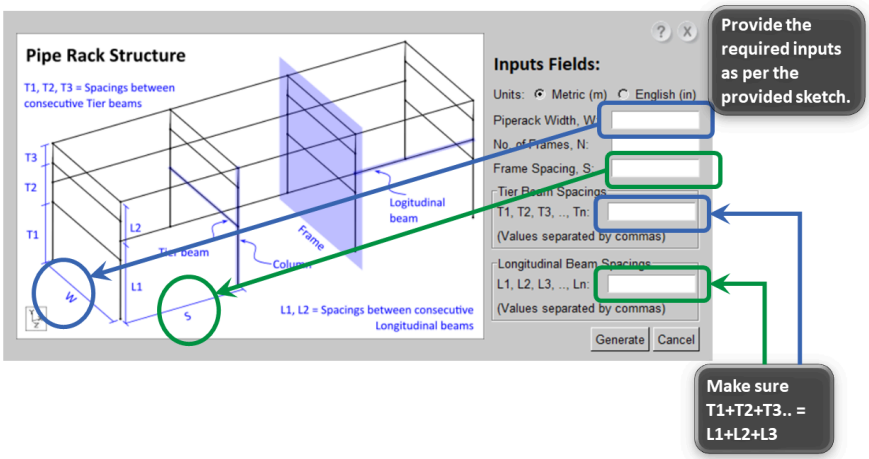
Pipe Rack and PEBs (Structure Wizards)

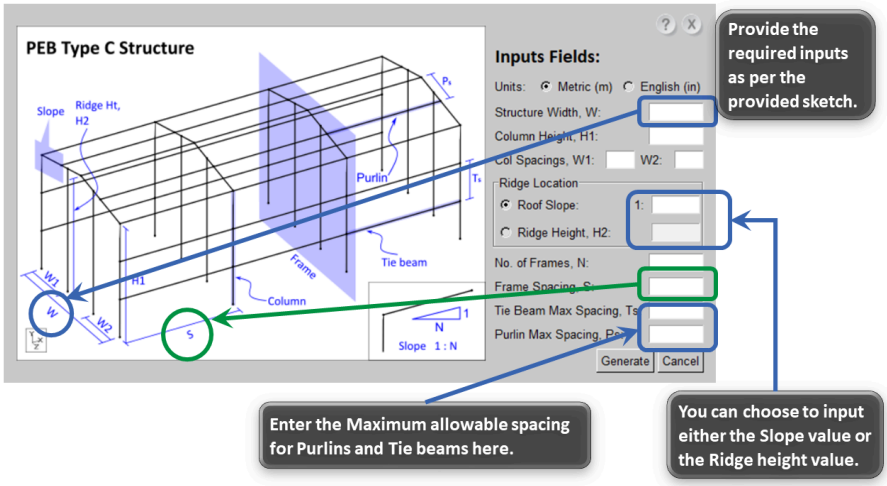


Keyboard shortcuts: Alt+P, Alt+A, Alt+B, Alt+C, Alt+D respectively

Create geometry for Pipe Racks or Pre-Engineered Buildings (PEB) with just a click.

- Click on any button of your choice: "**Pipe Rack**", "**PEB - A**", "**PEB - B**", "**PEB - C**" or "**PEB - D**".
- Simply provide the required inputs in the corresponding dialog boxes, following the provided sketch.
- Click the "**Generate**" button.
- This will create a new STAAD file with the Geometry based on your specified input dimensions.
- You can select either Metric or English units according to your preference.
- Detailed information for sample Geometry Generators is provided below:





Check our video guide on Structure Wizards

Complete Video Guide

Check our complete video guide featuring all tools